

Satyam Raj

+91 8529228967 | satyamraj1905@gmail.com | Portfolio Website |
LinkedIn | GitHub | LeetCode (960+ submissions) | CSES (250+ submissions)

EXPERIENCE

Software Developer

Sept 2025 – Present

Open Limits Pvt. Ltd. (Digital Streets)

Noida, India

- Developed and optimized production web applications with a performance-first approach, consistently meeting Core Web Vitals standards.
- Implemented performance improvements that led to faster load times, smoother interactions, and lower layout shift (LCP P75 ~1.7s, INP 64ms, CLS 0.01, +665% session growth, +46% revenue growth). [See detailed work report.](#)
- Collaborated with designers, marketers, and stakeholders and have experience in building multiple product price segment websites like luxury, professional, etc..

Research Fellow

June 2024 – July 2024

Indian Institute of Technology Kharagpur (IIT Kharagpur)

Kharagpur, India

- Selected for experimental validation of a biomedical antenna design at the Department of Electronics and Electrical Communication Engineering (ECE).
- Conducted antenna characterization using anechoic chamber and antenna test range facilities, including radiation pattern, gain, and impedance measurements.
- Analyzed measurement data against simulated results to evaluate real-world performance deviations and design robustness.

Research Intern

May 2023 – Nov 2023

LNMIIT, Jaipur

Jaipur, India

- Simulated multi-band biomedical antenna (ISM 2.4/5.1 GHz, MedRadio 900 MHz) using EM tools.
- Optimized SAR, gain, return loss and automated visualization using Python for 5x iteration efficiency.
- Documented 10+ performance metric comparisons and contributed to 2 technical drafts.

Teaching Assistant

Aug 2024 – Nov 2024

Digital Communication Lab, LNMIIT

Jaipur, India

- Guided 80+ students with 95% lab evaluation success in modulation, line coding, MATLAB tasks.
- Designed 8+ lab manuals and mini-assessments scripts to align with Bloom's taxonomy outcomes.

EDUCATION

LNMIIT, Jaipur

Jan 2022 – June 2025

B.Tech, Electronics and Communication Engineering

PROJECTS

[Doodle Chat](#) | React, Node.js, Socket.IO, Express

[Live Demo](#)

- Real-time drawing + chat app with 100ms latency WebSocket communication.
- Implemented dynamic room system with 10+ concurrent user handling.

[Lecture Hall Management System](#) | MERN Stack

[Live Demo](#)

- Role-based booking flow handling 4 user roles and 12+ endpoint APIs.
- Enabled auto-updating UI for booking statuses using Redux and WebSocket hooks.

[Parking Lot System](#) | Java (LLD)

[GitHub](#)

- OOP CLI system managing 3 vehicle types across 5 floors with scalable architecture.
- Thread-safe concurrent handling of up to 100 simulated parking operations.

TECHNICAL SKILLS

Languages: C/C++, HTML, CSS, Javascript, Typescript, Liquid, Java, Python (Micro)

Frontend: React.js, Recoil, Redux, Next.js, Tailwind CSS, npm

Backend: Node.js, Express.js, REST APIs, Git, GitHub

Database: MongoDB, Mongoose, MySQL, PostgreSQL, Prisma

Fundamentals: DSA (500+ problems), OOPS, DBMS, OS, CN

DevOps: AWS, Docker, Cloudflare

ACHIEVEMENTS

- Among the top 30 candidates whose project got approved at IIT KGP for practical scenario testing
- Secured INR 27k+ in 10+ inter-college sketch/photography events (IIT-KGP, IIT-D, St. Stephen's are some of the biggest names.).
- Earned INR 13k+ via 4+ commissioned sketches; [!\[\]\(97faa0168e491544be255cfcab218e9b_img.jpg\) view my work.](#)